TERTIARY INFOTECH ACADEMY

LAB 09
Troubleshoot a VPC Connectivity Path

AWS Certified SysOps Administrator Associate Training
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Scenario
A private application instance is healthy locally but cannot reach an approved service endpoint after a network change.

Objective
Apply K2 and A2 by isolating a failed request through DNS, routes, security groups, network ACLs, target health and VPC Flow Logs.

Deliverable
A VPC diagnostic worksheet, reachability/flow-log evidence, corrected network control and before/after result.

Services and tools
VPC · Reachability Analyzer · Flow Logs · Security Groups · NACLs · Endpoints

Safety and cost boundary
- Use only the trainer-authorised learner account and Region.
- Replace placeholders and tag learner-owned resources.
- Capture evidence before cleanup.
- Verify deletion and report retained chargeable resources.

Detailed procedure

Step 1 — Review `PATH-DIAGRAM.md`; write source, destination, protocol, port, expected route and success response.
```
open PATH-DIAGRAM.md
```

Evidence note: _____

Step 2 — Run `diagnostics.sh` with the learner source and destination; retain every command result even when it fails.
```
bash diagnostics.sh REPLACE_SOURCE_ENI REPLACE_DESTINATION_IP | tee evidence/diagnostics.txt
```

Evidence note: _____

Step 3 — Confirm DNS resolution and compare the returned address with the intended private/public path.
```
dig +short REPLACE_HOSTNAME | tee evidence/dns.txt
```

Evidence note: _____

Step 4 — Inspect source and destination subnet route tables using longest-prefix-match reasoning.
```
aws ec2 describe-route-tables --filters Name=association.subnet-id,Values=REPLACE_SUBNET_ID |
tee evidence/routes.json
```

Evidence note: _____

Step 5 — Inspect security-group rules in both directions and state whether return traffic is automatically allowed.

```
aws ec2 describe-security-groups --group-ids REPLACE_SG_ID | tee evidence/security-group.json
```

Evidence note: _____

Step 6 — Inspect network ACL rules by number in both directions, including ephemeral return ports.

```
aws ec2 describe-network-acls --filters Name=association.subnet-id,Values=REPLACE_SUBNET_ID |
tee evidence/nacl.json
```

Evidence note: _____

Step 7 — Create or run Reachability Analyzer where supported and record the first blocking component.

```
aws ec2 start-network-insights-analysis --network-insights-path-id REPLACE_PATH_ID | tee
evidence/reachability.json
```

Evidence note: _____

Step 8 — Query Flow Logs for ACCEPT/REJECT evidence correlated to source, destination, port and UTC time.

```
open FLOW-LOG-QUERY.txt
```

Evidence note: _____

Step 9 — Change only the identified learner-owned control; repeat the same test and record before/after evidence.

```
open DIAGNOSTIC-REPORT.md
```

Evidence note: _____

Step 10 — Capture route, rule and result as JPEG evidence; run validation and restore temporary changes.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence note: _____

Verification

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.

Acceptance criteria

- Source/destination/protocol/port explicit
- Route and policy evaluated separately
- Stateful/stateless return path explained
- One variable changed
- Before/after test is identical

Evidence checklist

□ DNS result

□ route tables

□ security-group and NACL exports

□ reachability analysis

□ Flow Log query

□ completed diagnostic report

□ JPEG evidence

□ cleanup.txt

Troubleshooting

- Check identity, Region and resource ID before changing permissions.
- Resolve every REPLACE placeholder.
- Preserve error output as evidence.
- Escalate cleanup failures immediately.